

more realistic animations. Wind creates a directional force, with additional controls for turbulence. Motor acts to turn rigid objects, such as the blades of a helicopter.

9.3.4 Constraints

Constraints let you restrict how objects move in the simulation. They can fix objects to each other or let them move only in specific ways. The Point-Point (point-to-point) constraint lets you attach two objects together. The objects can rotate in relation to each other, but always have the attachment point in common. The Point-Path constraint allows you to constrain two bodies so that the child can move along a path relative to the parent, much like a bead on a wire. The Prismatic constraint allows its bodies to move relative to each other along one axis only. The Ragdoll constraint lets you realistically simulate the behaviour of body joints, such as hips, shoulders, and ankles. This is very useful for creating digital stuntmen. You can use the Carwheel constraint to attach a wheel to another object—for instance, a car chassis. You can also constrain a wheel to a position in world space. During the simulation, the wheel object is free to rotate around a spin axis defined in each object's space. The spring constraint lets you create a spring like effect between two rigid bodies in the simulation, or between a rigid body and a point in space. The Hinge constraint allows you to simulate a hinge like action between two bodies.

9.3.5 Helpers

Helpers let you define specific types of simulations, such as an automobile or a fracturing object. Fracture simulates the breaking of a rigid body into pieces as the result of an impact. To do this, you need to create the pieces that are glued together to create the whole object. Toy Car simulates a simple car without having to set up each constraint separately. This option lets you choose a chassis and wheels for your car, tweak various properties such as the strength of its suspension, and specify whether you would like reactor to turn its wheels during the simulation. Soft bodies can be used to make this character's belly jiggle.

9.3.6 Helpers Soft Bodies

Soft bodies are objects that change shape when affected by forces and motion. They are great for secondary motion such as a waving flag, a jiggling belly, or a dog's floppy ears. Soft bodies are similar to cloth, the main difference